

Representations of the symmetric group

Math 737 - Lab 8

Exercise 1: For $n=3,4,5,6$, use the function `'.irreducible_characters()'` to show that the number of irreducible characters of S_n is the number of partitions of n .

```
In [8]: S=[]
I=[]
for i in range(3,7):
    S = SymmetricGroup(i)
    print(f"It is {len(S.irreducible_characters())==Partitions(i).cardinalit
```

It is True that the number of partitions of 3 = the number of irreducible characters of S_3 .

It is True that the number of partitions of 4 = the number of irreducible characters of S_4 .

It is True that the number of partitions of 5 = the number of irreducible characters of S_5 .

It is True that the number of partitions of 6 = the number of irreducible characters of S_6 .

Exercise 2: The code below does the following for $n=3$. Do the same for $n=4,5$.

- Compute the character table of S_n (including the conjugacy class representatives indexing the columns).
- By looking at the output and using your knowledge of character tables, state which partition corresponds to each row.

```
In [ ]: S3 = SymmetricGroup(3)
display(S3.conjugacy_classes_representatives())
display(S3.character_table())
print("The partitions indexing the rows (in order) are:",[1,1,1],[2,1],[3])
```

[(), (1,2), (1,2,3)]

[1 -1 1]

[2 0 -1]

[1 1 1]

The partitions indexing the rows (in order) are: [1, 1, 1] [2, 1] [3]

```
In [15]: S4 = SymmetricGroup(4)
display(S4.conjugacy_classes_representatives())
display(S4.character_table())
print("The partitions indexing the rows (in order) are:", [1,1,1,1], [2,1,1],
[()], (1,2), (1,2)(3,4), (1,2,3), (1,2,3,4)]
[ 1 -1  1  1 -1]
[ 3 -1 -1  0  1]
[ 2  0  2 -1  0]
[ 3  1 -1  0 -1]
[ 1  1  1  1  1]
The partitions indexing the rows (in order) are: [1, 1, 1, 1] [2, 1, 1] [2,
2] [3, 1] [4]
```

```
In [16]: S5 = SymmetricGroup(5)
display(S5.conjugacy_classes_representatives())
display(S5.character_table())
print("The partitions indexing the rows (in order) are:", [1,1,1,1,1], [2,1,
[()], (1,2), (1,2)(3,4), (1,2,3), (1,2,3)(4,5), (1,2,3,4), (1,2,3,4,5)]
[ 1 -1  1  1 -1 -1  1]
[ 4 -2  0  1  1  0 -1]
[ 5 -1  1 -1 -1  1  0]
[ 6  0 -2  0  0  0  1]
[ 5  1  1 -1  1 -1  0]
[ 4  2  0  1 -1  0 -1]
[ 1  1  1  1  1  1  1]
The partitions indexing the rows (in order) are: [1, 1, 1, 1, 1] [2, 1, 1,
1] [2, 2, 1] [3, 1, 1] [3, 2] [4, 1] [5]
```

Exercise 3: The code below does the following for $n=3$. Do the same for $n=4,5$.

- Check that the row orthogonality theorem holds.

```
In [ ]: irr = S3.irreducible_characters()
#irr is the list of irreducible characters. Since it is a list, we can access
display(irr[0].values())
display(irr[1].values())
display(irr[2].values())

[1, -1, 1]
[2, 0, -1]
[1, 1, 1]
```

```
In [21]: #Now use .scalar_product to perform the inner product of one character with
display(irr[0].scalar_product(irr[0]))
display(irr[0].scalar_product(irr[1]))
display(irr[0].scalar_product(irr[2]))

display(irr[1].scalar_product(irr[0]))
display(irr[1].scalar_product(irr[1]))
display(irr[1].scalar_product(irr[2]))
```

```
display(irr[2].scalar_product(irr[0]))
display(irr[2].scalar_product(irr[1]))
display(irr[2].scalar_product(irr[2]))
#Need to check the scalar product for all combinations of irr[i], irr[j] for
```

```
1
0
0
0
1
0
0
0
1
```

For S_4, S_5 I show it in matrix form where $M = (\langle \chi_i, \chi_j \rangle) = (\delta_{ij}) = I$

S_4

```
In [31]: irr = S4.irreducible_characters()

for i in range(len(irr)):
    display(irr[i].values())

print("-----")

M = matrix([
    [irr[i].scalar_product(irr[j]) for j in range(len(irr))]
    for i in range(len(irr))
])

display(M)
```

```
[1, -1, 1, 1, -1]
[3, -1, -1, 0, 1]
[2, 0, 2, -1, 0]
[3, 1, -1, 0, -1]
[1, 1, 1, 1, 1]
-----
[1 0 0 0 0]
[0 1 0 0 0]
[0 0 1 0 0]
[0 0 0 1 0]
[0 0 0 0 1]
```

S_5

```
In [32]: irr = S5.irreducible_characters()

for i in range(len(irr)):
    display(irr[i].values())

print("-----")

M = matrix([
```

```

    irr[i].scalar_product(irr[j]) for j in range(len(irr))]
    for i in range(len(irr))
])

display(M)

```

```

[1, -1, 1, 1, -1, -1, 1]
[4, -2, 0, 1, 1, 0, -1]
[5, -1, 1, -1, -1, 1, 0]
[6, 0, -2, 0, 0, 0, 1]
[5, 1, 1, -1, 1, -1, 0]
[4, 2, 0, 1, -1, 0, -1]
[1, 1, 1, 1, 1, 1, 1]
-----
[1 0 0 0 0 0 0]
[0 1 0 0 0 0 0]
[0 0 1 0 0 0 0]
[0 0 0 1 0 0 0]
[0 0 0 0 1 0 0]
[0 0 0 0 0 1 0]
[0 0 0 0 0 0 1]

```

Exercise 4: The code below does the following for $n=3$. Do the same for $n=4,5$.

- Sum the squares of the dimensions of each irreducible representation to obtain the order of the group.

```

In [39]: #Here is the calculation for n=3. For higher n, you should write a loop ratl
irr = S3.irreducible_characters()
list(irr[0])[0]^2+list(irr[1])[0]^2+list(irr[2])[0]^2

```

Out[39]: 6

The below cells show that we get $|S_4|=24=4!$, $|S_5|=5!=120$, respectively, as expected, when summing the squares of the dimensions of each irreducible representation.

```

In [40]: S4 = SymmetricGroup(4)
irr = S4.irreducible_characters()

sum(list(irr[i])[0]^2 for i in range(len(irr)))

```

Out[40]: 24

```

In [41]: S5 = SymmetricGroup(5)
irr = S5.irreducible_characters()

sum(list(irr[i])[0]^2 for i in range(len(irr)))

```

Out[41]: 120

